

Uptake of doublet and triplet therapy for men with *de novo* metastatic castration sensitive prostate cancer. Population-based study

Paolo Zaurito^{a, b}, Rolf Gedeberg^a, Hans Garmo^a, Eugenio Ventimiglia^a, Mats Alhberg^a, Pär Stattin^a, Marcus Westerberg^a

^aDepartment of Surgical Sciences, Uppsala University, Uppsala, Sweden
^bDivision of Experimental Oncology/Unit of Urology, Urological Research Institute, IRCCS Ospedale San Raffaele, Milan, Italy



UniSR
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INTRODUCTION

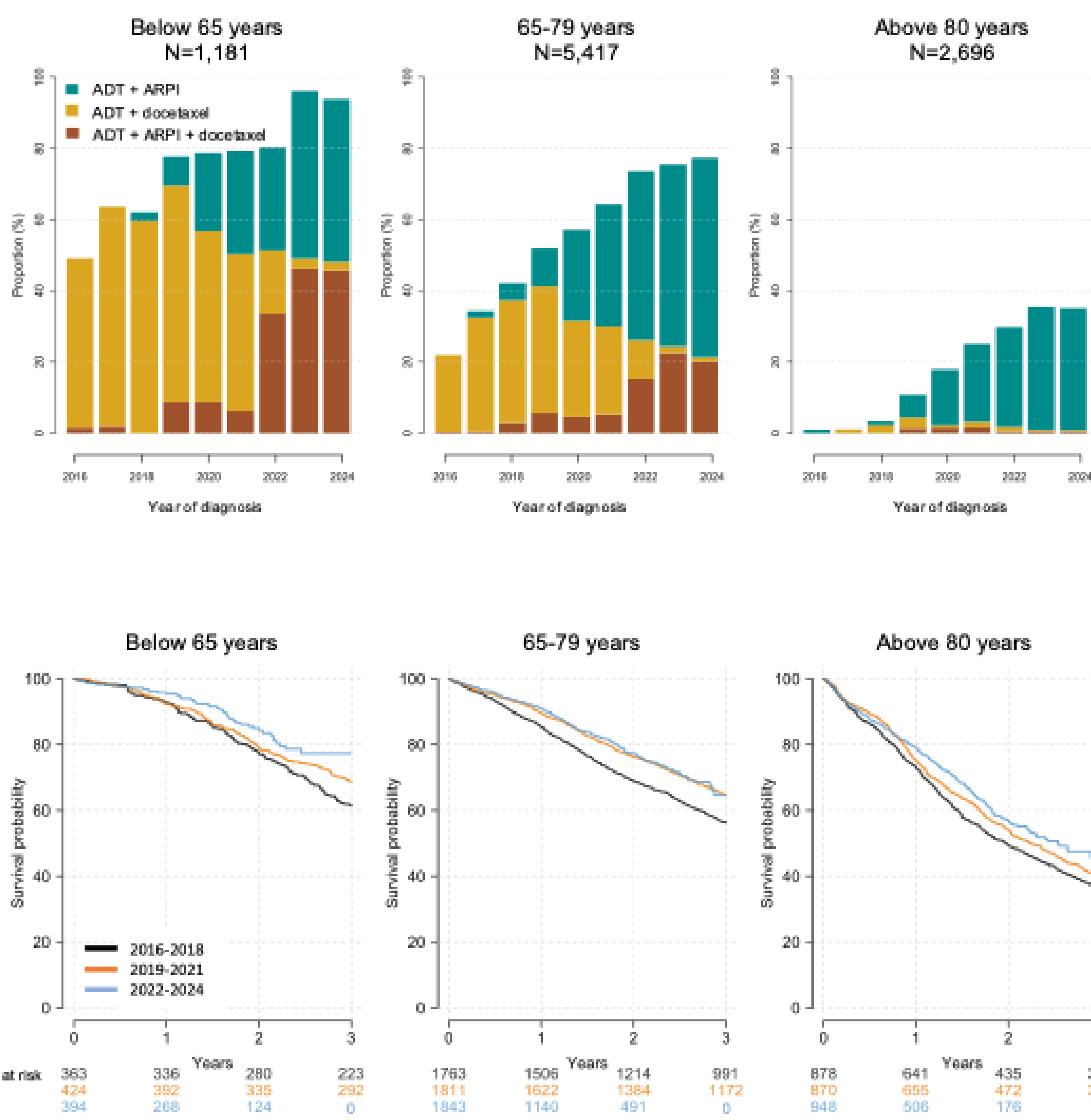
- In randomized clinical trials doublet and triplet therapy improved survival compared to standard androgen deprivation therapy (ADT) in men with *de novo* metastatic castration-sensitive prostate cancer (**mCSPC**).
- Guidelines recommend **doublet therapy since 2020** and **triplet therapy since 2022**.
- The aim of this study was to assess the uptake of upfront doublet and triplet therapy at a population level and assess trends in survival for all men with mCSPC.

MATERIALS AND METHODS

- We included men registered with **de novo mCSPC** in **2016-2024** in the National Prostate Cancer Register (**NPCR**) of Sweden.
- Use of **Docetaxel** within 6 months after diagnosis was extracted from both NPCR and regional health care IT systems. Use of androgen receptor pathway inhibitors (**ARPIs**) was based on filled prescriptions for abiraterone, enzalutamide, apalutamide and darolutamide in the Prescribed Drug Register.
- We estimated the annual proportion of men with *de novo* mCSPC who upfront received doublet therapy (ADT plus ARPI or docetaxel) or triplet therapy (ADT plus docetaxel and ARPI).
- Kaplan-Meier curves were used to estimate 3-year **overall survival** in three calendar periods (2016-2018, 2019-2021, and 2022-2024).

RESULTS

- Median PSA at diagnosis decreased slightly from 104 ng/mL (IQR: 28-434) in 2016-2018 to 89 ng/mL (IQR: 25-354) in 2022-2024.
- In **9,294** men diagnosed with *de novo* mCSPC the use of upfront **doublet therapy** increased from **19% in 2016** to **66% in 2024**, and use of **triplet therapy** rose from **4% in 2021** to **17% in 2024**.
- Men treated with triplet therapy were younger and had a more advanced disease, reflected by **high PSA levels, clinical T stage and Gleason score**.
- Three-year survival increased from **51% (95% CI: 49-52%)** in 2016-2018 to **61% (95% CI: 58-64%)** in 2022-2024. Among men below age 65, survival increased from 69% (95% CI: 65-73) in 2019-2021 to 77% (95% CI: 71-84) in 2022-2024.



CONCLUSION

The uptake of doublet and triplet therapy increased substantially during the study period, in particular among men below age 65. In parallel, 3-year overall survival increased in all men diagnosed with *de novo* mCSPC. These data provide support for benefit of upfront doublet or triplet therapy also in routine clinical practice.